



DIGITAL PRODUCT PASSPORT VERTICAL · WHITE PAPER

Product Provenance Is Now a Legal Requirement. The Infrastructure Doesn't Exist Yet.

Append-only supply chain records and governed compliance automation
— live and registry-ready before the EU Battery Regulation deadline.

COLLABORATIVE INTELLIGENCE * PARTICIPATION

ABSTRACT

Unlike every other disruption described in SOCIII's vertical white paper series, the wave hitting product provenance is not speculative. It is already law.

The EU Battery Regulation (2023/1542) requires a Digital Product Passport for all industrial batteries and light mobility transport batteries sold in the European Union by February 2027. The Ecodesign for Sustainable Products Regulation extends this requirement to textiles, electronics, and construction products on a rolling timeline through 2030. Any company that manufactures or imports covered products into the EU market and cannot produce a compliant DPP will be blocked from that market.

Feb 2027

EU BATTERY REGULATION
DPP DEADLINE — NO GRACE
PERIOD

1.2B

BATTERY UNITS/YEAR
AFFECTED BY THE
REGULATION

\$99/mo

SOCIII BASE PRICE VS. \$2–5M
FOR A CONSULTING
ASSESSMENT ALONE

WAVE 1 — THE EU REGULATORY MANDATE IS ALREADY LAW

NOW — FEBRUARY 2027

The EU Battery Regulation 2023/1542 entered into force on August 17, 2023. The DPP requirement for industrial batteries (>2kWh) takes effect February 18, 2027. The DPP requirement for light mobility transport (LMT) batteries — e-bikes, e-scooters, electric mobility devices — takes effect August 18, 2027. There is no opt-out, no phase-in grace period, and no small-business exemption for EU market access.

A compliant DPP requires: battery identification, performance and durability data, a carbon footprint declaration by lifecycle stage, material composition (including critical raw materials like cobalt, lithium, and nickel), supply chain due diligence documentation, end-of-life and recyclability data, and any incident and safety record. Every data point must be machine-readable, accessible via a QR code, and hosted on a system interoperable with the EU's central DPP registry.

This is not a trend to monitor. It is a compliance deadline to meet. The Ecodesign for Sustainable Products Regulation (ESPR) extends the same framework to textiles (2025), then electronics, construction products, furniture, and industrial goods.

WAVE 2 — SUPPLY CHAIN TRANSPARENCY LAWS ARE CONVERGING GLOBALLY

NOW — 3 YEARS

The EU is not acting alone. Germany's Supply Chain Act (LkSG) has required due diligence since January 2023. The EU Corporate Sustainability Due Diligence Directive (2024) extends similar requirements to all large EU companies and their non-EU suppliers. In the US, the Uyghur Forced Labor Prevention Act creates a rebuttable presumption that goods from Xinjiang involve forced labor, and the SEC's climate disclosure rules require large public companies to disclose Scope 3 emissions.

These laws converge on a single requirement: companies must produce a documented, traceable, verifiable record of where products came from, what they contain, and how they were made — persistent, not reconstructed at audit time. This is architecturally identical to chain-of-title in real estate. **The architecture is the same. The pending patent application covers both.**

WAVE 3 — CONSUMER AND BRAND DEMAND FOR AUTHENTIC PROVENANCE IS ACCELERATING

12 – 36 MONTHS

The regulatory mandate creates the floor. Consumer demand creates the ceiling. The global luxury goods market is \$400 billion/year; the counterfeit luxury goods market is estimated at \$30–50 billion/year. Every luxury brand is actively exploring authentication infrastructure, facing the same problem regulators face:

current provenance claims are self-asserted, paper-based, or locked in proprietary systems that cannot be verified by third parties.

The model that works is simple: a QR code on the product links to an append-only provenance record that is hosted, verified, and queryable through a standard interface. The manufacturer, the distributor, the retailer, and the end consumer all see the same record. No one can modify it retroactively.

WHY THE SOCIII ARCHITECTURE IS THE RIGHT FOUNDATION

Most DPP solutions proposed are one of two things: a centralized database managed by a compliance vendor with ongoing access fees, or a blockchain implementation that is technically decentralized but practically unusable because write costs are unpredictable and the query interface requires specialized tooling. SOCIII's append-only record is neither.

By default, a cryptographic hash of each record is anchored to the Base blockchain for publicly verifiable tamper-evidence — but the record itself lives in a professional-grade append-only ledger, not in on-chain storage. Write costs are predictable. The query interface is standard. This is not a consumer crypto product. The blockchain provides the tamper-evidence layer; SOCIII provides the infrastructure.

Every product event — a supplier's raw material attestation, a carbon footprint calculation, a certification renewal, an end-of-life disposition — is written as an immutable log entry referencing the one before it, forming a chain. No one can go back and change what a supplier attested in 2024 when an audit arrives in 2027, because that entry is not a field that can be edited — it is a permanent event in the chain. The AI validates each new entry against the regulatory requirements before it becomes a record.

- **Immutable by design.** A carbon footprint attestation written in 2025 cannot be retroactively changed when the regulation tightens in 2027.
- **Model-agnostic.** The RAAS rules engine validates every data input regardless of which AI model produced it.
- **Portable.** A manufacturer starting with SOCIII for EU Battery Regulation compliance doesn't need a separate system when ESPR extends to their textile line.

THE SOCIII DPP STACK

WORKER	WHAT IT DOES	RECORD IT CREATES
DPP Compliance	EU Battery Reg compliance check — all required data fields validated	dpp-compliance-report/v1
Passport Builder	Generates the compliant DPP data package with QR-linkable identifier	dpp-passport/v1
Supply Chain Tracer	Documents supply chain provenance and sourcing certifications	supply-chain-bundle/v1
Registry Manager	Submits and manages records in the EU DPP registry	registry-record/v1
Lifecycle Monitor	Tracks battery state of health, incident reports, end-of-life events	lifecycle-alert/v1

Alex, the Chief of Staff, monitors compliance deadlines across the entire product catalog — alerting when a DPP needs updating, when a supplier's certification is expiring, or when a new product category falls under ESPR scope.

WHAT THIS COSTS — AND WHAT THE ALTERNATIVE COSTS

WHAT YOU PAY NOW	SOCIII
McKinsey/Big Four assessment	\$200–900/month all-in
Implementation engagement (\$3–10M)	Included
Internal compliance team (3–5 FTEs)	Same 2-person team, same output
Timeline: 12–18 months, deadline risk high	On track for Feb 2027 if started now

The third option — doing nothing — is market exclusion from the EU. For any company with meaningful EU revenue, the cost of non-compliance is not a fine. It is losing access to the world's largest single market.

BEFORE

A 2-person compliance team manually building and maintaining DPP records for a 50-SKU catalog across three disconnected tools. Assembly is manual, auditable only to the extent of what was saved.

AFTER

Same 2-person team. Passport Builder handles the catalog. Alex monitors all deadlines. The audit trail is always current and always tamper-evident.

COMING IN Q3 2026: DIRECT SHOPIFY AND ECOMMERCE INTEGRATION

For brands selling through Shopify, WooCommerce, or other ecommerce platforms, SOCIII will publish a native app and API connector that makes DPP generation part of the product listing workflow — no separate compliance workflow, no data re-entry, no manual QR code generation.

THE PILOT: A COMPLIANCE CONSULTING FIRM AND THE MEDICAL DEVICE SUPPLY CHAIN

SOCIII's first DPP pilot is with Volta Advisory, a consulting firm guiding a medical device manufacturer's EU Battery Regulation compliance program for portable medical device battery products — equipment that crosses both EU medical device regulation (MDR) and EU Battery Regulation scope simultaneously. This is the hardest version of the problem: two regulatory frameworks, multiple supply chain tiers, and a product where the provenance record has patient safety implications. The learnings translate directly to the broader EU Battery Regulation and ESPR market.

CONCLUSION

Product provenance documentation has been a voluntary best practice for decades. It is now mandatory law in the world's largest single market, with hard deadlines, clear penalties, and a regulatory trajectory that extends to most durable goods categories by 2030.

The wave is mandatory. The deadline is visible. The architecture exists. The price is \$99/month.

"I manufacture lithium batteries and I need to know which products need a DPP before February."

Alex reviews the catalog against the regulation's scope criteria — battery capacity, chemistry type, application category — and returns a compliance checklist ranked by urgency. Before you have opened a

spreadsheet, you know which products are in scope, which are borderline, and what documentation work is required for each.